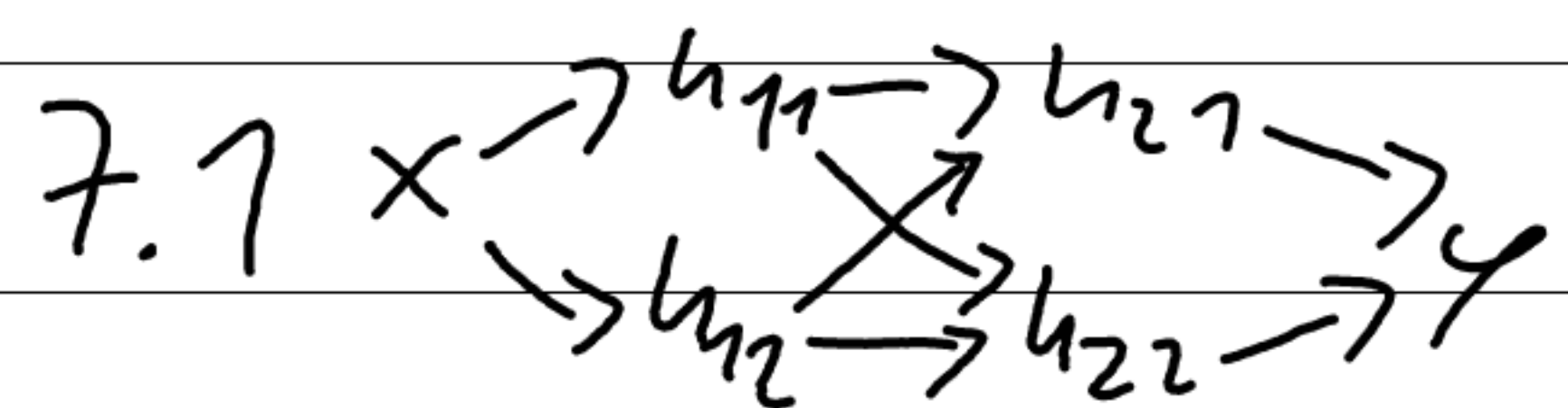


7 Gradients and initialization



$$\begin{aligned} \text{with } z_{11,\text{in}} &= \theta_{01} + \theta_{11}x, \quad z_{11,\text{out}} = \mathcal{Q}[z_{11,\text{in}}] \\ z_{12,\text{in}} &= \theta_{02} + \theta_{12}x, \quad z_{12,\text{out}} = \mathcal{Q}[z_{12,\text{in}}], \\ z_{21,\text{in}} &= \psi_{01} + \psi_{11}z_{11,\text{out}} + \psi_{21}z_{12,\text{out}}, \\ z_{22,\text{in}} &= \psi_{02} + \psi_{12}z_{11,\text{out}} + \psi_{22}z_{12,\text{out}}, \\ z_{21,\text{out}} &= \mathcal{Q}[z_{21,\text{in}}], \quad z_{22,\text{out}} = \mathcal{Q}[z_{22,\text{in}}] \end{aligned}$$

$$y = \phi_0 + \phi_1 z_{1,out} + \phi_2 z_{2,out}$$

$$\frac{\partial \varphi}{\partial \phi_0} = 1 \quad \frac{\partial \varphi}{\partial \phi_1} = z_{21, \text{out}} \quad \frac{\partial \varphi}{\partial \phi_2} = z_{22, \text{out}}$$

$$\frac{\partial y}{\partial \psi_{01}} = \phi_1 I[\bar{z}_{21,in} > 0] \quad \frac{\partial y}{\partial \psi_{02}} = \phi_2 I[\bar{z}_{22,in} > 0]$$

$$\frac{\partial \varphi}{\partial \varphi_{17}} = \phi_7 I[z_{27, in} > 0] \cdot z_{17, out} \quad \frac{\partial \varphi}{\partial \varphi_{27}} = \phi_7 I[z_{27, in} > 0] \cdot z_{12, out}$$

$$\frac{\partial \psi}{\partial \psi_{12}} = \phi_2 I[\bar{z}_{22, in} > 0] \cdot z_{11, out} \quad \frac{\partial \psi}{\partial \psi_{22}} = \phi_2 I[\bar{z}_{22, in} > 0] z_{12, out}$$

$$\frac{\partial \psi}{\partial \theta_{01}} = \phi_1 I[z_{21, in} > 0] \psi_{11} I[z_{11, in} > 0] + \phi_2 I[z_{22, in} > 0] \psi_{12} I[z_{11, out} > 0]$$

$$\frac{\partial \varphi}{\partial \theta_{02}} = \phi_1 I[\varepsilon_{21,i1} > 0] \psi_{21} I[\varepsilon_{12,i1} > 0] + \phi_2 I[\varepsilon_{22,i1} > 0] \psi_{22} I[\varepsilon_{12,i1} > 0]$$

$$\frac{\partial \psi}{\partial \theta_{17}} = \phi_1 \mathbb{I}[\bar{z}_{27,in} > 0] \psi_{11} \mathbb{I}[\bar{z}_{11,in} > 0] x + \phi_2 \mathbb{I}[\bar{z}_{22,in} > 0] \psi_{12} \mathbb{I}[\bar{z}_{12,out} > 0] x$$

$$\frac{\delta \phi}{\delta \theta_{12}} = \phi_7 I[\tau_{27,i_1} > 0] \psi_{27} I[\tau_{72,i_1} > 0] x + \phi_2 I[\tau_{22,i_1} > 0] \psi_{22} I[\tau_{27,i_1} > 0] x$$

$$7.2 \quad \frac{\partial l_i}{\partial p_3} = 2(p_3 - y_i) \quad \frac{\partial p_3}{\partial h_3} = w_3 \quad \frac{\partial h_3}{\partial p_2} = -\sin(p_2)$$

$$\frac{\partial p_2}{\partial h_2} = w_2 \quad \frac{\partial h_2}{\partial p_1} = \exp(p_1) \quad \frac{\partial p_1}{\partial h_1} = w_1$$

$$\frac{\partial h_1}{\partial p_0} = \cos(p_0)$$

$$\Rightarrow \frac{\partial l_i}{\partial p_2} = 2(p_3 - y_i) w_3 (-\sin(p_2))$$

$$\frac{\partial l_i}{\partial h_2} = w_2 \frac{\partial l_i}{\partial p_2}$$

$$\frac{\partial l_i}{\partial p_1} = \exp(p_1) \frac{\partial l_i}{\partial h_2}$$

$$\frac{\partial l_i}{\partial h_1} = w_1 \frac{\partial l_i}{\partial p_1}$$

$$\frac{\partial l_i}{\partial p_0} = \cos(p_0) \frac{\partial l_i}{\partial h_1}$$

$$7.3 \quad \frac{\partial l_i}{\partial p_0} = \frac{\partial h_1}{\partial p_0} \frac{\partial p_1}{\partial h_1} \frac{\partial h_2}{\partial p_1} \frac{\partial p_2}{\partial h_2} \frac{\partial h_3}{\partial p_2} \frac{\partial p_3}{\partial h_3} \frac{\partial l_i}{\partial p_3}$$

$$D_1 \times 1 = D_1 \times D_1 \quad D_1 \times D_2 \quad D_2 \times D_2 \quad D_2 \times D_3 \quad D_3 \times D_3 \quad D_3 \times D_1 \quad D_1 \times 1$$

$$7.4 \quad \frac{\partial \ell_i}{\partial f(x_i, \phi)} = -2(y_i - f(x_i, \phi)) \quad \text{or} \quad \ell_i = (y_i - f(x_i, \phi))^2$$

$$7.5 \quad \frac{\partial \ell_i}{\partial f(x_i, \phi)} = (1 - y_i)(1 - \text{sig}(f(x_i, \phi)))^{-1} \cdot \text{sig}(f(x_i, \phi)) \cdot (1 - \text{sig}(f(x_i, \phi)))^{-1} \\ - y_i (1 - \text{sig}(f(x_i, \phi)))^{-1} \cdot \text{sig}(f(x_i, \phi)) \cdot (1 - \text{sig}(f(x_i, \phi)))^{-1} \\ = \text{sig}(f(x_i, \phi)) - y_i$$

$$\text{as } \frac{\partial \text{sig}(x)}{\partial x} = \text{sig}(x) \cdot (1 - \text{sig}(x))$$

$$7.6 \quad z = \beta + \Omega h$$

$$\text{so: } \frac{\partial z}{\partial h} = \Omega^T$$

$$\frac{\partial z_i}{\partial h_j} = \frac{\partial}{\partial h} \left(\sum_k \Omega_{ik} h_k \right) = \Omega_{ij}$$

$$7.7 \quad \frac{\partial \text{sig}(f)}{\partial f} = \text{sig}(f) \cdot (1 - \text{sig}(f)) \quad (\text{from 5-DL})$$

$$f \rightarrow \infty \stackrel{a}{=} 1 \quad f \rightarrow -\infty \stackrel{a}{=} 0$$

$$7.8 \quad \text{Heaviside}(z) = \begin{cases} 0, & z < 0 \\ 1, & z \geq 0 \end{cases}$$

both are problematic
as their derivatives are
0 everywhere (step functions)

$$\text{rect}(z) = \begin{cases} 0 & z < 0 \\ 1 & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases}$$

7.9 $\ell(f)$ with $f = \beta + \Omega h$

$$h: \frac{\partial \ell}{\partial \Omega} = \frac{\partial \ell}{\partial f} h^T$$

$$\frac{\partial \ell}{\partial \Omega} = \frac{\partial \ell}{\partial f} \underbrace{\frac{\partial f}{\partial \Omega}}_{\frac{\partial}{\partial \Omega} \beta + \Omega h = h^T} = \frac{\partial \ell}{\partial f} h^T$$

$$7.10 \quad \frac{\partial \text{ReLU}(z)}{\partial z} = \begin{cases} \alpha & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$

$\leadsto$ backpropagation the same except update step:

$$\frac{\partial \ell_i}{\partial f_{k-1}} = \mathbb{I}[f_{k-1} > 0] \odot \left(\Omega_k^T \frac{\partial \ell_i}{\partial f_k} \right) + \mathbb{I}[f_{k-1} < 0] \odot \alpha \left(\Omega_k^T \frac{\partial \ell_i}{\partial f_k} \right)$$

7.11 Basic idea: $\leadsto$ save every batch layers pre-activations ($\hat{=}$ checkpoint)

2 $\leadsto$ recompute a block from last checkpoint

3 $\leadsto$ backprop that block

4 $\leadsto$ discard

5 $\leadsto$ move on

layers are
from
checkpoint
to the next

$$7.12 \quad y = \exp[\exp[x] + \exp[x]^2] + \sin[\exp[x] + \exp[x]^2]$$

$$f_1 = \exp[x]$$

$$f_2 = f_1^2$$

$$f_3 = f_1 + f_2$$

$$f_4 = \exp[f_3]$$

$$f_5 = \sin[f_3]$$

$$y = f_4 + f_5$$

$$\frac{\partial y}{\partial f_5} = 1$$

$$\frac{\partial y}{\partial f_4} = 1$$

$$\frac{\partial y}{\partial f_3} = \frac{\partial y}{\partial f_4} \frac{\partial f_4}{\partial f_3} + \frac{\partial y}{\partial f_5} \frac{\partial f_5}{\partial f_3} = \exp[f_3] + \cos[f_3]$$

$$\frac{\partial y}{\partial f_2} = \frac{\partial y}{\partial f_3} \frac{\partial f_3}{\partial f_2} = \exp[f_3] + \cos[f_3]$$

$$\begin{aligned} \frac{\partial y}{\partial f_1} &= \frac{\partial y}{\partial f_2} \frac{\partial f_2}{\partial f_1} + \frac{\partial y}{\partial f_3} \frac{\partial f_3}{\partial f_1} \\ &= (\exp[f_3] + \cos[f_3]) 2f_1 \\ &\quad + \exp[f_3] + \cos[f_3] \end{aligned}$$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial f_1} \frac{\partial f_1}{\partial x}$$

$$= ((\exp[f_3] + \cos[f_3]) 2f_1 + \exp[f_3] + \cos[f_3]) \exp[x]$$

$$7.13 \quad \frac{\partial f_1}{\partial x} = \exp[x] \quad \frac{\partial f_2}{\partial x} = \frac{\partial f_2}{\partial f_1} \frac{\partial f_1}{\partial x} = 2f_1 \exp[x]$$

$$\frac{\partial f_3}{\partial x} = \frac{\partial f_3}{\partial f_1} \frac{\partial f_1}{\partial x} + \frac{\partial f_3}{\partial f_2} \frac{\partial f_2}{\partial x} = \exp[x] + 2f_1 \exp[x]$$

$$\frac{\partial f_4}{\partial x} = \frac{\partial f_4}{\partial f_3} \frac{\partial f_3}{\partial x} = \exp[f_3] (\exp[x] + 2f_1 \exp[x])$$

$$\frac{\partial f_5}{\partial x} = \frac{\partial f_5}{\partial f_3} \frac{\partial f_3}{\partial x} = \cos[f_3] (\exp[x] + 2f_1 \exp[x])$$

$$\begin{aligned} \frac{\partial y}{\partial x} &= \frac{\partial y}{\partial f_4} \frac{\partial f_4}{\partial x} + \frac{\partial y}{\partial f_5} \frac{\partial f_5}{\partial x} = \exp[f_3] \exp[x] + 2f_1 \exp[f_3] \exp[x] \\ &\quad + \cos[f_3] \exp[x] + 2f_1 \cos[f_3] \exp[x] \end{aligned}$$

↳ forward-mode diff.
scale with number of inputs:
one pass per parameter

backward-mode diff
needs one pass per
all parameters (as loss scalar)

7.14 motivation: when activations pass through layers and nonlinearities (ReLU), we want their scale (variance $\hat{=}$ 2nd moment or $E(x^2)$) to stay in a healthy range (neither explode nor vanish)

Let a be zero-mean RV with symmetric distribution ($a \stackrel{d}{=} -a$),
 $b = \text{ReLU}(a)$.

$$\text{or } E[b^2] = \frac{\sigma^2}{2} \text{ where } \sigma^2 = \text{Var}(a)$$

$$b^2 = a^2 I[a \geq 0]$$

Let $p(x)$ density of a . As a symmetric with zero mean,
 $p(-x) = p(x)$. Thus $E[p(x) I[a \geq 0]] = \frac{1}{2} E[p(x)]$

$$E[b^2] = E[a^2 I[a \geq 0]] = \frac{1}{2} E[a^2]$$

as $\text{Var}(a) = E[a^2] - E[a]^2 = E[a^2]$ as $E[a] = 0$

$$E[b^2] = \frac{1}{2} \text{Var}(a) = \frac{\sigma^2}{2} \quad \square$$

7.15 if we used the ReLU activation, the network would be 'stuck' with these zero weights & biases as
e.g. gradient descent would equal zero in every step!

with smooth activations (e.g., tanh, sigmoid), signals can flow, but they receive identical gradients and remain identical forever !!